Indices, Surds and Polynomials

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Indices, Surds & Polynomials Practice Exercises**

Express $\frac{3+\sqrt{5}}{2-\sqrt{5}}$ in the form $p+q\sqrt{5}$, where p and q are rational numbers to be found.

Multiply numerator and denominator by the conjugate of the denominator, $2+\sqrt{5}$:
$\frac{3+\sqrt{5}}{2-\sqrt{5}} \times \frac{2+\sqrt{5}}{2+\sqrt{5}} = \frac{(3+\sqrt{5})(2+\sqrt{5})}{2^2-(\sqrt{5})^2}$ Expanding the numerator: $3(2)+3\sqrt{5}+2\sqrt{5}+(\sqrt{5})^2 = 6+5\sqrt{5}+5 = 11+5\sqrt{5}$.
The denominator is $4-5=-1$. Therefore: $\frac{11+5\sqrt{5}}{-1} = -11-5\sqrt{5}$ Thus $p=-11$ and $q=-5$.

2 **[Foundational]** • **Target: Indices, Surds & Polynomials Practice Exercises**

[About A2 Level · Practice 2] If the governing relation for this system is $g(t)=4t+4$, evaluate $g(6)$ and find the inverse input t when $g(t)=32$.

Substitute $t=6$: $g(6)=4(6)+4=\mathbf{28}$. Setting $4t+4=32$ gives $4t=28 \Rightarrow t=\mathbf{7}$.

3 **[Foundational]** • **Target: Indices, Surds & Polynomials Practice Exercises**

[Algebra: Partial Fractions & Division · Practice 3] If the governing relation for this system is $g(t)=5t+5$, evaluate $g(8)$ and find the inverse input t when $g(t)=50$.

Substitute $t=8$: $g(8)=5(8)+5=\mathbf{45}$. Setting $5t+5=50$ gives $5t=45 \Rightarrow t=\mathbf{9}$.

4 **[Foundational]** • **Target: Indices, Surds & Polynomials Practice Exercises**

[Complex Numbers · Practice 3] If the governing relation for this system is $g(t)=5t+5$, evaluate $g(8)$ and find the inverse input t when $g(t)=50$.

Substitute $t=8$: $g(8)=5(8)+5=\mathbf{45}$. Setting $5t+5=50$ gives $5t=45 \Rightarrow t=\mathbf{9}$.

5 **[Foundational]** • **Target: Indices, Surds & Polynomials Practice Exercises**

[Continuous Random Variables · Practice 3] If the governing relation for this system is $g(t)=5t+5$, evaluate $g(8)$ and find the inverse input t when $g(t)=50$.

Substitute $t=8$: $g(8)=5(8)+5=\mathbf{45}$. Setting $5t+5=50$ gives $5t=45 \Rightarrow t=\mathbf{9}$.

6 **[Foundational]** • **Target: Indices, Surds & Polynomials Practice Exercises**

[Differentiation · Practice 3] If the governing relation for this system is $g(t)=5t+5$, evaluate $g(8)$ and find the inverse input t when $g(t)=50$.

Substitute $t=8$: $g(8)=5(8)+5=\mathbf{45}$. Setting $5t+5=50$ gives $5t=45 \Rightarrow t=\mathbf{9}$.

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7 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

[Exponentials & Logarithms · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Indices, Surds & Polynomials Practice Exercises

[Hypothesis Testing · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

9 [Intermediate] • Target: Indices, Surds & Polynomials Practice Exercises

[Integration · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

10 [Intermediate] • Target: Indices, Surds & Polynomials Practice Exercises

[Numerical Methods · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

11 [Intermediate] • Target: Indices, Surds & Polynomials Practice Exercises

[Poisson Distribution · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

12 [Intermediate] • Target: Indices, Surds & Polynomials Practice Exercises

[Sampling & Estimation · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

13 [Intermediate] • Target: Indices, Surds & Polynomials Practice Exercises

[Sequences & Series · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

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14 **[Intermediate]** • **Target: Indices, Surds & Polynomials Practice Exercises**

[Advanced Trigonometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Indices, Surds & Polynomials Practice Exercises** • **Cross-Lesson: 4.01 Logarithmic and Exponential Functions**

[Vectors · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

16 **[Challenging]** • **Target: Indices, Surds & Polynomials Practice Exercises** • **Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure**

[Algebra & Graphs · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

17 **[Challenging]** • **Target: Indices, Surds & Polynomials Practice Exercises** • **Cross-Lesson: 6.01 Permutations and Combinations**

[Geometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

18 **[Challenging]** • **Target: Indices, Surds & Polynomials Practice Exercises** • **Cross-Lesson: 7.01 Differentiation and Rates of Change**

[Mensuration · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

Substitute $t = 6$: $g(6) = 4(6) + 4 = \mathbf{28}$. Setting $4t + 4 = 32$ gives $4t = 28 \Rightarrow t = \mathbf{7}$.

19 **[Challenging]** • **Target: Indices, Surds & Polynomials Practice Exercises** • **Cross-Lesson: 8.01 Integration and Kinematics**

[Number · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

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20 [Challenging] • Target: Indices, Surds & Polynomials Practice Exercises • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

[Sets & Venn Diagrams · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

Find the remainder when the polynomial $f(x) = 4x^3 - 6x^2 + 3x - 1$ is divided by $2x - 1$.

a. $-\frac{1}{2}$ ✓

b. $\frac{1}{2}$

c. $-\frac{9}{2}$

d. 0

By the Remainder Theorem, when a polynomial $f(x)$ is divided by a linear factor of the form $ax - b$, the remainder is $f(\frac{b}{a})$. Here, the divisor is $2x - 1$, so we set $2x - 1 = 0 \Rightarrow x = \frac{1}{2}$. Substitute $x = \frac{1}{2}$ into $f(x)$: $f(\frac{1}{2}) = 4(\frac{1}{2})^3 - 6(\frac{1}{2})^2 + 3(\frac{1}{2}) - 1 \ = 4(\frac{1}{8}) - 6(\frac{1}{4}) + \frac{3}{2} - 1$.

22 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

When the polynomial $\mathrm{P}(x) = 2x^3 - 3x^2 + ax - 8$ is divided by $x - 3$, the remainder is 10. What is the value of the constant a?

a. -3 ✓

b. 3

c. -9

d. -33

According to the Remainder Theorem, when a polynomial $\mathrm{P}(x)$ is divided by $x - c$, the remainder is equal to $\mathrm{P}(c)$. Here, the divisor is $x - 3$, so $c = 3$. We evaluate $\mathrm{P}(3)$: $\mathrm{P}(3) = 2(3)^3 - 3(3)^2 + a(3) - 8$ $\mathrm{P}(3) = 2(27) - 3(9) + 3a - 8 = 54 - 27 + 3a - 8 = 19 + 3a$ Given that the remainder is 10: $19 + 3a = 10$ $3a = 10 - 19 = -9$ $a = -3$ Common misconceptions: - Choosing 3 results from a sign error when rearranging $19 + 3a = 10$ to $3a = 19 - 10 = 9$..

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23 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

The polynomial $p(x) = 2x^3 + ax^2 - 5x + 6$ is divisible by $x - 2$. Find the value of the constant a.

a. -3 ✓

b. 3

c. 0

d. -6

By the Factor Theorem, if $p(x)$ is divisible by $x - 2$, then $p(2) = 0$. Substitute $x = 2$ into the polynomial expression: $p(2) = 2(2)^3 + a(2)^2 - 5(2) + 6$ $p(2) = 2(8) + 4a - 10 + 6 = 16 + 4a - 10 + 6 = 4a + 12$ Setting $p(2) = 0$: $4a + 12 = 0 \Rightarrow 4a = -12 \Rightarrow a = -3$ Common misconceptions: - Using the root $x = -2$ instead of $x = 2$ leads to $2(-8) + 4a - 5(-2) + 6 = 4a = 0 \Rightarrow a = 0$. - A sign error when solving the linear equation $4a + 12 = 0$ incorrectly gives $a = 3$.

24 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

In the expansion of $\left(1 + \frac{x}{2}\right)^n$, the coefficient of x^2 is twice the coefficient of x. Given $n > 0$, what is the value of n?

a. **9** ✓

b. 11

c. 8

d. 5

Coeff of x is $\frac{n}{2}$. Coeff of x^2 is $\frac{n(n-1)}{2(4)}$. Given $\frac{n(n-1)}{2 \cdot 4} = 2\left(\frac{n}{2}\right) \Rightarrow \frac{n-1}{2 \cdot 2} = 2 \Rightarrow n - 1 = 4(2) = 8 \Rightarrow n = 9$.

25 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

The polynomial $P(x) = 2x^2 + ax - 4$ is exactly divisible by $(x - 4)$. What is the value of the constant a?

a. -10.00

b. 7.00

c. **-7.00** ✓

d. -5.00

By the Factor Theorem, $P(4) = 0 \Rightarrow 2(4)^2 + a(4) - 4 = 0 \Rightarrow 32 + 4a - 4 = 0 \Rightarrow a = -7.00$.

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26 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

The function f is defined for all real numbers x by $f(x) = 2 - 3\sin(4x)$. What are the amplitude and the period (in radians) of f?

a. Amplitude $= 3$, Period $= \frac{\pi}{2}$ ✓

b. Amplitude $= -3$, Period $= \frac{\pi}{2}$

c. Amplitude $= 3$, Period $= 8\pi$

d. Amplitude $= 5$, Period $= \frac{\pi}{4}$

For a sinusoidal function written in the form $f(x) = c + a\sin(bx)$ (or $c - a\sin(bx)$): 1. **Amplitude:** The amplitude represents the maximum vertical distance from the horizontal centerline (midline $y = 2$) to a peak or trough. It is defined as the absolute value of the coefficient of the trigonometric ratio: Amplitude $= |-3| = 3$. Amplitude is strictly a non-negative measure of distance; it can never be negative. 2. **Period:** The fundamental period of $\sin(x)$ is 2π. For $\sin(bx)$.

27 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

In triangle ABC, $AB = 6$ cm, $AC = 10$ cm, and the area of triangle ABC is $15\sqrt{3}$ cm^2. Given that angle BAC is obtuse, find the exact length of BC.

a. 14 cm ✓

b. $2\sqrt{19}$ cm

c. $2\sqrt{34}$ cm

d. 8 cm

1. Use the formula for the area of a triangle: Area $= \frac{1}{2}(AB)(AC)\sin(\angle BAC)$ $15\sqrt{3} = \frac{1}{2}(6)(10)\sin(\angle BAC) = 30\sin(\angle BAC)$ $\sin(\angle BAC) = \frac{15\sqrt{3}}{30} = \frac{\sqrt{3}}{2}$ 2. Since $\angle BAC$ is obtuse ($90^\circ < \angle BAC < 180^\circ$): $\angle BAC = 180^\circ - 60^\circ = 120^\circ$ 3. Determine $\cos(\angle BAC)$: $\cos(120^\circ) = -\cos(60^\circ) = -\frac{1}{2}$ 4. Apply the cosine rule to find BC.

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28 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

The variables x and y are related by the equation $y = 2x^3 - 5x$. Given that x is increasing at a constant rate of 3 units per second, find the rate of change of y with respect to time at the instant when $x = 2$.

a. 57 units per second ✓

b. 19 units per second

c. $\frac{19}{3}$ units per second

d. 21 units per second

To find the rate of change of y with respect to time, apply the chain rule for connected rates of change: $\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$ 1. Differentiate y with respect to x: $\frac{dy}{dx} = \frac{d}{dx}(2x^3 - 5x) = 6x^2 - 5$ 2. Evaluate $\frac{dy}{dx}$ at $x = 2$: $\frac{dy}{dx}\Big|_{x=2} = 6(2)^2 - 5 = 6(4) - 5 = 24 - 5 = 19$ 3. Multiply by.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Indices, Surds & Polynomials Practice Exercises

The variables x and y are connected by the equation $y = 2x^3 - 5x$. Given that x increases at a constant rate of 4 units s^{-1}, find the rate of change of y at the instant when $x = 2$.

a. 76 units s^{-1} ✓

b. 19 units s^{-1}

c. 28 units s^{-1}

d. 24 units s^{-1}

To find the rate of change of y with respect to time t, we apply the chain rule for connected rates of change: $\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$ First, differentiate y with respect to x: $\frac{dy}{dx} = \frac{d}{dx}(2x^3 - 5x) = 6x^2 - 5$ Substitute $x = 2$ into the derivative: $\frac{dy}{dx}|_{x=2} = 6(2)^2 - 5 = 24 - 5 = 19$ Given that.

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30 [Intermediate] • Target: Indices, Surds & Polynomials Practice Exercises

Find the real solution to the equation $\log_3 x + \log_3 (x-8) = 2$.

a. $x = 9$ ✓

b. $x = 9$ and $x = -1$

c. $x = 8.5$

d. $x = 5$

Apply the product rule of logarithms, $\log_a u + \log_a v = \log_a (uv)$: $\log_3 [x(x-8)] = 2$ Convert the logarithmic equation to index (exponential) form: $x(x-8) = 3^2$ $x^2 - 8x = 9$ $x^2 - 8x - 9 = 0$ Factor the quadratic: $(x-9)(x+1) = 0$ which gives potential solutions $x = 9$ or $x = -1$. Check the domain of the original logarithmic expression: for $\log_3 x$ and $\log_3 (x-8)$ to be defined for real numbers, we must have $x > 0$ and $x - 8 > 0$, which requires $x > 8$. Thus, $x = -1$ is an extra.

31 [Intermediate] • Target: Indices, Surds & Polynomials Practice Exercises

A curve has equation $y = \frac{kx}{2x-1}$, where $x eq \frac{1}{2}$ and k is a non-zero constant. The normal to the curve at the point where $x = 1$ has a gradient of $\frac{1}{3}$. Given that x is increasing at a rate of 0.6 units per second, find the rate of change of y when $x = 2$.

a. -0.2 units per second ✓

b. 0.2 units per second

c. -1.8 units per second

d. -0.6 units per second

Step 1: Find the derivative $\frac{dy}{dx}$ in terms of k using the quotient rule. Let $u = kx$ and $v = 2x - 1$. Then $u' = k$ and $v' = 2$. $\frac{dy}{dx} = \frac{k(2x-1) - 2(kx)}{(2x-1)^2} = \frac{2kx - k - 2kx}{(2x-1)^2} = \frac{-k}{(2x-1)^2}$ Step 2: Find the parameter k. At $x = 1$, the gradient of the tangent is: $m_T = \frac{dy}{dx}\Big|_{x=1} = \frac{-k}{(2(1)-1)^2} = -k$ The gradient of the normal is given by $m_N = -\frac{1}{m_T} = \frac{1}{k}$. Since the gradient of the normal is.

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32 [Intermediate] • Target: Indices, Surds & Polynomials Practice Exercises

At 14:00, a coastguard radar station detects two vessels, A and B. Relative to the radar station, vessel A has position vector $(-4\mathbf{i} + 6\mathbf{j})$ km and moves with a constant velocity of $(8\mathbf{i} + 5\mathbf{j})$ km h^{-1}. Vessel B has position vector $(20\mathbf{i} - 4\mathbf{j})$ km and moves with a constant velocity of $(u\mathbf{i} + 10\mathbf{j})$ km h^{-1}, where u is a constant. Given that vessel B intercepts vessel A, find the value of u.

a. $u = -4$ ✓

b. $u = 4$

c. $u = -2$

d. $u = -8$

Let t be the time in hours after 14:00 when the interception occurs. The position vector of vessel A at time t is: $\mathbf{r}_A(t) = (-4 + 8t)\mathbf{i} + (6 + 5t)\mathbf{j}$ The position vector of vessel B at time t is: $\mathbf{r}_B(t) = (20 + ut)\mathbf{i} + (-4 + 10t)\mathbf{j}$ For interception to occur, $\mathbf{r}_A(t) = \mathbf{r}_B(t)$. Equating the $\mathbf{j}$-components: $6 + 5t = -4 + 10t \Rightarrow 5t = 10 \Rightarrow t = 2$ hours Equating the $\mathbf{i}$-components and substitu.

33 [Intermediate] • Target: Indices, Surds & Polynomials Practice Exercises

The variables x and y satisfy the relationship $y = Ab^x$, where A and b are positive constants. The graph of $\ln y$ against x is a straight line passing through the points $(0, \ln 3)$ and $(4, \ln 48)$. What is the value of b?

a. 2 ✓

b. 4

c. 16

d. $\sqrt{2}$

Taking the natural logarithm of both sides of the equation $y = Ab^x$: $\ln y = \ln(Ab^x) = \ln A + x\ln b$ This is in linear form $Y = mX + c$, where $Y = \ln y$, $X = x$, the gradient is $m = \ln b$, and the vertical intercept is $c = \ln A$. From the point $(0, \ln 3)$, the intercept is $\ln A = \ln 3$, which gives $A = 3$. Using the second point $(4, \ln 48)$: $\ln 48 = \ln 3 + 4\ln b$ Subtract $\ln 3$ from both sides and apply the quotient rule of logarithms.

34 [Intermediate] • Target: Indices, Surds & Polynomials Practice Exercises

The position vectors of points A and B relative to an origin O are given by $\vec{OA} = 5\mathbf{i} - 2\mathbf{j}$ and $\vec{OB} = -\mathbf{i} + 6\mathbf{j}$. Find the magnitude of $\vec{AB}$.

a. $4\sqrt{2}$

b. **10** ✓

c. $2\sqrt{13}$

d. 14

To find the vector $\vec{AB}$, subtract the position vector of A from that of B: $\vec{AB} = \vec{OB} - \vec{OA} = (-\mathbf{i} + 6\mathbf{j}) - (5\mathbf{i} - 2\mathbf{j}) = (-1 - 5)\mathbf{i} + (6 - (-2))\mathbf{j} = -6\mathbf{i} + 8\mathbf{j}$ Next, calculate the magnitude $|\vec{AB}|$: $|\vec{AB}| = \sqrt{(-6)^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10$ Common misconceptions: - Calculating $|\vec{OA} + \vec{OB}|$ gives $|4\mathbf{i} + 4\mathbf{j}| = \sqrt{16 + 16} = 4\sqrt{2}$. - Making a sign error when sub.

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35 [Intermediate] • Target: Indices, Surds & Polynomials Practice Exercises

Solve the logarithmic equation $\log_2(5x+6) - \log_2(x) = 3$.

a. $x = 2$ ✓

b. $x = 6$

c. $x = \frac{1}{2}$

d. $x = -3$

1. Apply the logarithm quotient law $\log_a(A) - \log_a(B) = \log_a\left(\frac{A}{B}\right)$: $\log_2\left(\frac{5x+6}{x}\right) = 3$ 2. Convert the equation from logarithmic to exponential form: $\frac{5x+6}{x} = 2^3 = 8$ 3. Multiply through by x (noting $x > 0$): $5x + 6 = 8x \Rightarrow 3x = 6 \Rightarrow x = 2$ 4. Checking the domain: Both $5(2) + 6 = 16 > 0$ and $2 > 0$, so $x = 2$ is a valid solution. Common misconceptions: - Calculating $2^3 = 6$ instead of 8 yields $5x + 6 = 6x \Rightarrow x = 6$..

36 [Intermediate] • Target: Indices, Surds & Polynomials Practice Exercises

The position vectors of points A and B relative to an origin O are $\vec{OA} = 2\mathbf{i} - 3\mathbf{j}$ and $\vec{OB} = 7\mathbf{i} + 9\mathbf{j}$. Find the unit vector in the direction of $\vec{AB}$.

a. $\frac{1}{13}(5\mathbf{i} + 12\mathbf{j})$ ✓

b. $\frac{1}{13}(-5\mathbf{i} - 12\mathbf{j})$

c. $\frac{1}{\sqrt{61}}(5\mathbf{i} + 6\mathbf{j})$

d. $\frac{1}{15}(9\mathbf{i} + 12\mathbf{j})$

First, find the displacement vector $\vec{AB}$: $\vec{AB} = \vec{OB} - \vec{OA} = (7\mathbf{i} + 9\mathbf{j}) - (2\mathbf{i} - 3\mathbf{j}) = (7-2)\mathbf{i} + (9-(-3))\mathbf{j} = 5\mathbf{i} + 12\mathbf{j}$. Next, calculate the magnitude of $\vec{AB}$: $|\vec{AB}| = \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$. The unit vector in the direction of $\vec{AB}$ is: $\frac{\vec{AB}}{|\vec{AB}|} = \frac{1}{13}(5\mathbf{i} + 12\mathbf{j})$. - Choice B gives the unit vector for.

37 [Intermediate] • Target: Indices, Surds & Polynomials Practice Exercises

Find the value of the integer $n \geq 3$ that satisfies the equation ${}^{n}\mathrm{P}_3 = 10 \times {}^{n}\mathrm{C}_2$

a. $n = 7$ ✓

b. $n = 12$

c. $n = 5$

d. $n = 8$

Using the algebraic definitions of permutations and combinations: ${}^{n}\mathrm{P}_3 = \frac{n!}{(n-3)!} = n(n-1)(n-2)$ ${}^{n}\mathrm{C}_2 = \frac{n!}{2!\,(n-2)!} = \frac{n(n-1)}{2}$ Substitute these into the given equation: $n(n-1)(n-2) = 10 \times \frac{n(n-1)}{2}$ $n(n-1)(n-2) = 5n(n-1)$ Since $n \geq 3$, we have $n(n-1)e0$. Dividing both sides by $n(n-1)$ gives: $n - 2 = 5 \Rightarrow n = 7$ Common misconceptions: - Forgetting the $2!$ denominator in ${}^{n}\mathrm{C}_2$ (treat.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Indices, Surds & Polynomials Practice Exercises • Cross-Lesson: 7.01 Differentiation and Rates of Change

Given that ${}^nP_2 = n3$, where n is an integer such that $n \geq 3$, find the value of n.

a. 8 ✓

b. 5

c. 3

d. 6

Using the algebraic definitions of permutations and combinations: ${}^nP_2 = \frac{n!}{(n-2)!} = n(n-1)$ and $\binom{n}{3} = \frac{n!}{3!\,(n-3)!} = \frac{n(n-1)(n-2)}{3!} = \frac{n(n-1)(n-2)}{6}$ Setting them equal gives: $n(n-1) = \frac{n(n-1)(n-2)}{6}$ Since $n \geq 3$, $n(n-1)e0$. Dividing both sides by $n(n-1)$: $1 = \frac{n-2}{6}$ Multiplying by 6: $6 = n-2 \Rightarrow n = 8$ Common misconceptions: - Confusing $3!$ with 3 leads to $n-2 = 3 \Rightarrow n = 5$. - Forgetting the division by $3!$.

39 [Challenging] • Target: Indices, Surds & Polynomials Practice Exercises • Cross-Lesson: 8.01 Integration and Kinematics

A particle moves in a straight line such that its acceleration, a m s^{-2}, at time t seconds is given by $a = 6t - 4$. When $t = 1$, the velocity of the particle is 5 m s^{-1}. Find the velocity of the particle when $t = 3$.

a. 15 m s^{-1}

b. 19 m s^{-1}

c. 21 m s^{-1} ✓

d. 27 m s^{-1}

To find the velocity v, integrate the acceleration function with respect to t: $v = \int a\,dt = \int (6t-4)\,dt = 3t^2 - 4t + C$ Using the initial condition that $v = 5$ when $t = 1$: $3(1)^2 - 4(1) + C = 5 \Rightarrow -1 + C = 5 \Rightarrow C = 6$ Thus, the velocity function is: $v(t) = 3t^2 - 4t + 6$ Substitute $t = 3$: $v(3) = 3(3)^2 - 4(3) + 6 = 27 - 12 + 6 = 21$ m s^{-1} Common misconceptions: - Omitting the constant of integration C gives.

40 [Challenging] • Target: Indices, Surds & Polynomials Practice Exercises • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

The position vectors of points A and B relative to a fixed origin O are given by $\vec{OA} = -2\mathbf{i} + 5\mathbf{j}$ and $\vec{OB} = 4\mathbf{i} - 3\mathbf{j}$. Find the magnitude of the vector $\vec{AB}$.

a. **10** ✓

b. $2\sqrt{2}$

c. 14

d. $2\sqrt{5}$

To find the vector $\vec{AB}$, subtract the position vector of A from the position vector of B: $\vec{AB} = \vec{OB} - \vec{OA} = (4\mathbf{i} - 3\mathbf{j}) - (-2\mathbf{i} + 5\mathbf{j}) = (4 - (-2))\mathbf{i} + (-3 - 5)\mathbf{j} = 6\mathbf{i} - 8\mathbf{j}$. The magnitude of $\vec{AB}$ is calculated using the Pythagorean theorem: $|\vec{AB}| = \sqrt{6^2 + (-8)^2} = \sqrt{36 + 64} = \sqrt{100} = 10$. Common misconceptions include adding the position vectors rather than subtracting them, giving.

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41 [Challenging] • Target: Indices, Surds & Polynomials Practice Exercises • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

A sector of a circle of radius 12 cm has a central angle of $\frac{\pi}{3}$ radians. Find the perimeter of the sector, giving your answer in exact form.

a. $(24 + 4\pi)$ cm ✓

b. $(12 + 4\pi)$ cm

c. 4π cm

d. $(24 + 24\pi)$ cm

To find the perimeter of a sector, we add the arc length to the two bounding radii. 1. Calculate the arc length s: $s = r\theta = 12 \times \frac{\pi}{3} = 4\pi$ cm 2. Calculate the perimeter P: $P = 2r + s = 2(12) + 4\pi = 24 + 4\pi$ cm Analysis of incorrect choices: - Choice B: Adds only one radius to the arc length, $r + s = 12 + 4\pi$. - Choice C: Computes only the arc length $s = 4\pi$, failing to include the two bounding radii. - Choice D: Incorrectly adds the full circumference.

42 [Challenging] • Target: Indices, Surds & Polynomials Practice Exercises • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

A sector of a circle of radius 12 cm has an arc length of 18 cm. Find the area of the sector.

a. 108 cm^2 ✓

b. 216 cm^2

c. 54 cm^2

d. 144 cm^2

The arc length s of a sector of radius r subtending an angle θ radians at the centre is given by $s = r\theta$. Substituting the given values gives $18 = 12\theta \Rightarrow \theta = 1.5$ radians. The area A of the sector is given by $A = \frac{1}{2}r^2\theta = \frac{1}{2}(12)^2(1.5) = 108$ cm^2 (or directly via $A = \frac{1}{2}rs = \frac{1}{2} \times 12 \times 18 = 108$ cm^2). Omitting the factor of $\frac{1}{2}$ yields the incorrect value of 216 cm^2, while.

43 [Challenging] • Target: Indices, Surds & Polynomials Practice Exercises • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

Given that ${}^nP_3 = 8\,{}^nC_2$, find the value of the integer n, where $n \geq 3$.

a. 6 ✓

b. 8

c. 10

d. 12

By definition of permutations and combinations: ${}^nP_3 = \frac{n!}{(n-3)!} = n(n-1)(n-2)$ and ${}^nC_2 = \frac{n!}{2!\,(n-2)!} = \frac{n(n-1)}{2}$. Substitute these expressions into the given equation: $n(n-1)(n-2) = 8\left(\frac{n(n-1)}{2}\right) = 4n(n-1)$. Since $n \geq 3$, $n(n-1)eq0$. Dividing both sides by $n(n-1)$ gives: $n - 2 = 4 \Rightarrow n = 6$. Common error: Omitting the factor of $2!$ in the denominator of ${}^nC_2$ leads to $n - 2 = 8 \Rightarrow n = 10$.

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44 [Challenging] • Target: Indices, Surds & Polynomials Practice Exercises • Cross-Lesson: 6.01 Permutations and Combinations

A particle moves in a straight line with acceleration $a(t) = 6t - 8$ m/s^2. If it starts from rest at the origin (at $t = 0$), at what time $t > 0$ does the particle momentarily come to rest again?

a. $t = 4$ s

b. $t = \frac{4}{3}$ s

c. $t = \frac{8}{3}$ s ✓

d. $t = 8$ s

Velocity $v(t) = \int (6t - 8)\,dt = 3t^2 - 8t + C$. Since $v(0) = 0 \Rightarrow C = 0$. Momentarily at rest when $v(t) = 0 \Rightarrow t(3t - 8) = 0 \Rightarrow t = \frac{8}{3}$ s.

45 [Challenging] • Target: Indices, Surds & Polynomials Practice Exercises • Cross-Lesson: 7.01 Differentiation and Rates of Change

Air is pumped into a spherical balloon such that its radius increases at a constant rate of 0.5 cm/s. What is the rate of change of volume $\frac{dV}{dt}$ when the radius is $r = 3$ cm?

a. 18π cm^3/s ✓

b. 36π cm^3/s

c. 9π cm^3/s

d. $\frac{18}{2}\pi$ cm^3/s

$V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$. At $r = 3$, $\frac{dV}{dt} = 4\pi(3)^2(0.5) = 2\pi(9) = 18\pi$ cm^3/s.

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